

Reply to Referee Report

Manuscript JHEP_249P_0226

On the Quantum Spectral Curve for $\text{AdS}_3 \times \text{S}^3 \times \text{S}^3 \times \text{S}^1$ strings and the $\mathfrak{d}(2,1;\alpha)$ Q -system

March 26, 2026

We are very grateful to the Referee for their thorough and careful reading of our manuscript, and for their positive assessment of our work. We are pleased that they found our results original and the presentation of a high standard, and we thank them for recommending publication in JHEP.

Below we address each of the Referee's comments in turn. We have revised the manuscript accordingly, and we believe these changes further improve the clarity of the paper.

Comment 1.

Page 22, below eq. (3.9). The conclusion $G = \hat{G}$ does not follow from eq. (3.8) alone — it additionally requires that G be u -independent, which should be stated explicitly at this point.

Reply:

We have added a line at page 22 that makes more explicit our assumption of G and $\hat{G}$ being u -independent.

Comment 2.

Page 3, bullet point 2. The claim that equivalence with Tsuboi's Q -system is “completely proved” in Appendix A is a minor overstatement, since Appendix A.2 acknowledges that the precise definition of the required untwisting limit is left to future work. This should be qualified in the introduction.

Reply:

We have corrected our claim on page 3, bullet point 2. We now point out that the two Q -systems structurally coincide after some natural assumption on the existence of an untwisting limit.

Comment 3.

Asymmetry between L and R wings: The L wing uses Dynkin diagram 1b while the R wing uses diagram 1a. The authors should clarify whether this asymmetry is physical or merely a gauge choice.

Reply:

We have added a comment and a footnote at page 47 that clarifies this choice for the gradings of the Bethe equations.

Comment 4.

Constraint 2, eq. (2.24). The argument that g_{ij} is constant uses the assumption that Q -functions have no singularities in either half-plane. It is not clear this applies to S and $\hat{S}$ in the full QSC, which have infinitely many short branch cuts. The authors should clarify whether Constraint 2 remains valid in this setting.

Reply:

We have modified the text under equations (2.27)-(2.28) refining our argument - in particular, we clarify that for g_{ij} to be constant it is only necessary that S and $\hat{S}$ are free of singularities either in the upper or lower half-plane, which is the same analyticity structure appearing in the full QSC.

We hope that the revised manuscript fully addresses the referee's comments.

With best regards,

Andrea Cavaglià, Rouven Frassek, Nicolò Primi, Roberto Tateo.